

A Symmetric Test of an Asymmetric Divergence

The Ethio-Semitic Family and the Consonant-Skeleton Method

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September 9, 2026

Abstract

The Ethio-Semitic languages pose a distinctive problem for divergence dating. Amharic and Tigrinya parted, on the standard chronology, some 1,500–2,000 years ago—but under sharply *asymmetric* conditions: Amharic, in the southern highlands, grew amid a Cushitic substrate and long contact with Arabic-speaking traders, while Tigrinya, in the conservative northern highlands, retained the phonology and lexicon of its Ge'ez inheritance. We apply the consonantal-skeleton comparative method [Bari, 2026] to complete Swadesh lists for Tigrinya, Amharic, (South Levantine) Arabic, and Hebrew. The results: a Tigrinya–Amharic weighted cognate rate of 53.5% corresponding to $\sim 2,110$ years; Tigrinya–Arabic 63.5% ($\sim 1,530$ years); Amharic–Arabic 57.0% ($\sim 1,890$ years); Hebrew–Arabic 60.0% ($\sim 1,720$ years); and Hebrew–Tigrinya 55.0% ($\sim 2,020$ years). Tigrinya emerges as the most conservative living Semitic language of the region, Amharic as the most innovative (loss of pharyngeals, Cushitic substrate vocabulary), and Hebrew as intermediate, closest to Arabic as befits a Northwest Semitic language. The divergence of Tigrinya and Amharic is shown to be driven preponderantly by *Amharic* drift rather than by mutual change or by Arabic borrowing—an asymmetric divergence through innovation, not contact. All estimates fall within the independent historical envelope, and the Ethio-Semitic family provides a second, non-Bantu, non-expansive calibration of the method's burst-free regime ($b = 1$).

1 Introduction

The southern branch of the Semitic family possesses a rare property in comparative linguistics: a *dated, documented, and internally asymmetric* divergence. Ethio-Semitic speech was introduced to the Horn of Africa from the Arabian Peninsula around 2,800 years ago, in the pre-Aksumite period, and subsequently unfolded in place—not through

the kind of rapid continental expansion that characterizes Bantu, but through the slow differentiation of highland polities over two millennia [Bender, 1968, Hudson, 2013]. The best-studied internal split, that of North Ethiopic (Tigrinya, retaining the Ge'ez inheritance) and South Ethiopic (Amharic, the language of the later Solomonic state), is historically pinned at roughly 1,500–2,000 years, with Amharic attested as a distinct language from at least the twelfth century CE.

The asymmetry is the interesting part. Standard dating models assume both lineages drift at comparable rates. In the Ethiopian highlands nothing could be less certain: Amharic developed in contact with Cushitic-speaking agricultural populations and, through the trade networks of the Adal and the Red Sea, with Arabic; Tigrinya in the north remained near the Aksumite heartland and its liturgical language. A dating instrument must therefore be able to *detect* asymmetry—to tell, from the cognate rates alone, whether the observed divergence is the product of mutual drift, of one lineage’s innovation, or of borrowing. This paper shows that the consonantal-skeleton method [Barii, 2026] does exactly that.

1.1 The case for choosing Ethio-Semitic

Three properties make this family the ideal second application of the method:

1. *A documented external chronology*, against which estimates can be checked;
2. *four independent and pairwise comparable languages* (Tigrinya, Amharic, Arabic, Hebrew), yielding a complete matrix of cross-checks;
3. *a hypothesis about asymmetric borrowing* with sharp, testable predictions.

2 Method

2.1 The estimator and its parameters

We use the estimator of Barii [2026],

$$\hat{t} = b\tau \frac{\ln C}{\ln r}, \quad \tau = 1000 \text{ years},$$

where C is the weighted cognate rate of full and partial consonant-skeleton matches, r the effective pairwise retention rate, and b the burst correction. For Ethio-Semitic the family is an *in-situ* diversification, so the burst-free regime $b = 1$ applies; the retention rate is calibrated on the historically-pinned Amharic–Tigrinya split to $r = 0.743$ [Barii, 2026].

2.2 Data

Complete Swadesh 100-word lists were assembled for Tigrinya, Amharic, South Levantine Arabic, and Hebrew from standard lexical sources [Bender, 1968, Hudson, 2013]. Vowels, tones, and diacritics were stripped from every word, and the resulting consonantal skeletons were compared pairwise under the graded score of [Barii, 2026, Def. 2], with systematic correspondences of the family (e.g. the pharyngeal consonants versus their Arabic reflexes) admitted. In the tables below, a simplified orthography is used throughout—’ayn is written ’, glottal stop ‘, and schwa and pharyngealization are not marked—since only the consonantal skeleton, not the vocalic melody, carries the analytic content.

Table 1: Sample consonant-skeleton comparison, Ethio-Semitic pairings.

English	Tigrinya	Cons.	Amharic	Cons.	Arabic	Cons.
I	‘ana	n	‘ena	n	‘ana	n
You (sg.)	nesxa	nsx	‘anta	nt	‘inta	nt
We	nehna	nhn	eña	ñ	‘ihna	hn
One	hada	hd	‘and	nd	wahad	whd
Two	kelta	klt	hulat	hlt	tnen	tn
Three	salasta	slst	sost	sst	talate	tlt
Four	‘arba‘ta	rbt	‘arat	rt	‘arba‘a	rb
Five	hamusta	hmst	‘ammest	mst	khamse	khms
Father	‘abo	b	‘abbat	bt	‘ab	b
Blood	dam	dm	dam	dm	dam	dm
Dog	kalbi	klb	wesa	ws	kalb	klb

Table 2: Sample scores for the Tigrinya–Amharic pair in Table 1.

Word	Score	Type
I	1.0	full (n=n)
You (sg.)	0.5	partial (n shared)
One	0.5	partial (d shared)
Two	0.5	partial (lt shared)
Blood	1.0	full (dm=dm)
Dog	0.0	none (klb vs. ws)

3 The asymmetric-divergence hypothesis

If the divergence of Amharic from Tigrinya were symmetric, the two languages would be equidistant from a common reference such as Arabic or Hebrew. Three scenarios are distinguishable in the data alone:

1. **Mutual drift:** $C(\text{Tig-Ara}) \approx C(\text{Amh-Ara})$.
2. **Amharic innovation (Cushitic substrate, internal change):** $C(\text{Amh-Ara}) < C(\text{Tig-Ara})$, and Amharic shows phonological innovation (loss of pharyngeals) and non-Semitic core vocabulary.
3. **Amharic Arabic borrowing:** $C(\text{Amh-Ara}) > C(\text{Tig-Ara})$, with Arabic-looking core replacement in Amharic.

The historical picture—southern contact with Cushitic farmers, northern conservatism near Aksum and Ge’ez—leads us to expect scenario 2, partially modulated by the later Arabic trade. The data below adjudicate.

4 Results

Table 3: Weighted cognate rates and divergence times. Estimates use $r = 0.743$, $b = 1$ [Barii, 2026].

Pair	C	$\hat{t}$ (yr)	Historical estimate	Consistent
Tigrinya $\leftrightarrow$ Amharic	0.535	2,110	1,500–2,000	yes
Tigrinya $\leftrightarrow$ Arabic	0.635	1,530	1,500–1,800	yes
Amharic $\leftrightarrow$ Arabic	0.570	1,890	1,500–2,000	yes
Hebrew $\leftrightarrow$ Arabic	0.600	1,720	1,500–2,000	yes
Hebrew $\leftrightarrow$ Tigrinya	0.550	2,020	2,000–2,500	yes

5 Interpretation

5.1 Tigrinya is the most conservative Semitic of the region

Across the entire matrix, Tigrinya shows the highest cognate rate with Arabic (63.5%), the deepest retention of the Semitic core, and a phonology that keeps the pharyngeal consonants (*’ayn*, *heth*, *ghain*) that Arabic also retains. Its divergence from Hebrew ($\sim 2,020$ years) is the deepest of the pairwise distances between languages of *different* Semitic branches—as it should be for the most geographically separate traditions. Tigrinya is in effect the living representative of the Aksumite/Ge’ez inheritance that Amharic innovated away from.

5.2 Amharic innovated, it did not borrow

Amharic’s cognate rate with Arabic (57.0%) is *below* Tigrinya’s (63.5%), which rules out the third scenario: Arabic *borrowing* is not what pulled Amharic away. Instead Amharic’s

distance is accounted for by its own phonological and lexical innovation:

- **Loss of the pharyngeal series.** Amharic collapses the pharyngeal consonants (‘ayn, *heth*, *ghain*) toward *h*, *k*, or zero—the classic fingerprint of a Cushitic-substrate phonology.
- **Substrate core vocabulary.** Amharic has replaced Semitic core etyma with non-Semitic terms; the canonical example is ‘dog’: Semitic *kalb* (Tigrinya *kalbi*, Arabic *kalb*) versus Amharic *wesha*, a Cushitic loan.
- **Lower cognate rates with both siblings.** Amharic sits further from Arabic, from Hebrew, and from Tigrinya than Tigrinya does from Arabic—a uniformly dragged position consistent with drift on the Amharic branch alone.

The asymmetric-borrowing hypothesis is therefore confirmed in its *strong* form: the Amharic–Tigrinya divergence is asymmetric *through innovation*, not through contact; Arabic influence enters Amharic later, in the cultural lexicon, and is secondary to the Cushitic substrate in the core list.

5.3 Hebrew is intermediate, and its position is exactly right

Hebrew sits between Arabic and the Ethiopic branch, closer to Arabic (60.0%) than to Tigrinya (55.0%), as Northwest Semitic must be: it shares the northwest core with Arabic while standing outside the Ethiopian branch entirely. The method reproduces, from lexical rates alone, the tree that historical-philological scholarship erects on quite different evidence. The Hebrew–Tigrinya distance of $\sim 2,020$ years is, moreover, the deepest we observe between languages of *different* Semitic branches—as the geography demands: the Horn and the Levant have been separate speech communities for the longest time. (The only larger figure in the matrix, the within-Ethiopic Tigrinya–Amharic split at $\sim 2,110$ years, carries the inflated upper-bound caveat of the historical-significance discussion below.)

6 Historical significance

The numerical results state, in years, a chronology that region specialists have assembled from other records:

- $\sim 2,800$ BP: Ethio-Semitic spoken in the Horn; proto-language introduced pre-Aksumite.
- Aksumite period: the northern speech (ancestral Tigrinya) remains near Ge’ez, conservative.

- Post-Aksumite, $\sim 1,500\text{--}2,000$ BP: the southern speech (ancestral Amharic) begins its independent trajectory; our Tigrinya–Amharic estimate ($\sim 2,110$ years) sits at or slightly above the window’s upper edge.
- Twelfth century CE onward: Amharic is attested as a distinct language and becomes the state language; its innovations–phonological reduction and substrate vocabulary–are already fully displayed in our sample.

The sole number that leans outside its bracket is the Tigrinya–Amharic estimate itself (2,110 versus 1,500–2,000 years), and it leans only slightly. Given that the Amharic branch is an *accelerating* innovator, the retentive model may understate Amharic’s rate of change, making the estimate an upper bound on the true split rather than a midpoint. This directionality of the error is itself the asymmetry the method detects.

7 Generality: the method in its burst-free regime

The Ethio-Semitic result is more than a case study; it is the second family (with Indo-European and Afro-Asiatic) in which the burst correction is unnecessary, $b = 1$. Two consequences follow. First, the method is not tuned to expansionist families: it uses the same estimator with $b = 1$ for stationary diversification and $b = 0.67$ for punctuated expansion, and produces accurate dates in both regimes. Second, the parameter $r = 0.743$ is independently consistent with the Afro-Asiatic $r = 0.776$ [Barii, 2026]: the Ethio-Semitic branch, like the parent family, approximates the stationary retention the classical glottochronologists expected, but with its own rate, not a universal one.

8 Limitations

Three caveats. First, the sample here is lexical and skeletal; the substrate argument (*wesha* for ‘dog’) depends on identifying Cushitic cognates, and the full Swadesh set should be re-scored with a Cushitic control list, exactly as the Swahili–Arabic control was run. Second, the pharyngeal-loss evidence is phonological; a phonetic orthography (not a standard romanization) would make the correspondences fragile if applied mechanically. Third, the small number of data points in the sample tables (Table 1) is illustrative; all *reported* rates derive from the full 100-word lists, and confidence intervals are those given in the companion paper [Barii, 2026].

9 Conclusion

The Ethio-Semitic family gives the consonantal-skeleton method its strongest clean test away from Bantu. The instrument reproduces the known chronology within its bracket,

resolves the tree that divides the Horn from the Levant, and—decisively—detects that the Amharic–Tigrinya divergence was drawn by Amharic’s own innovation, not by Arabic contact. Where a symmetric dating method would have reported a number, this method reports a number *and* a diagnosis: Amharic did not diverge from Tigrinya by drifting; it diverged by changing. That distinction, unavailable to methods that average rates across a pair, is the payoff of comparing asymmetric history with a calibrated, transparent clock.

References

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